

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
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Hands On 5 Implement Ansible Roles in Playbooks

1. PROCEDURE OUTPUT

Task 1: Use when command for different distributions

1. In the local machine, make sure you are in the local repository directory (CPE232_yourname). Issue the command `git pull`. When prompted, enter the correct passphrase or password. Describe what happened when you issued this command. Did something happen? Why?

It showed an output of 'Already up to date.' this because github uses content to track whether current versions match, and when I did a clone of the repository it is technically the same version because I have not done any commits and pushes yet.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ git pull
Already up to date.
qimagalan-workstation@workstation:~/CPE212_Galan$
```

2. Edit the inventory file and add the IP address of the Centos VM. Issue the command we used to execute the playbook (the one we used in the last activity): `ansible-playbook --ask-become-pass install_apache.yml`. After executing this command, you may notice that it did not become successful in the Centos VM. You can see that the Centos VM has failed=1. Only the two remote servers have been changed. The reason is that Centos VM does not support "apt" as the package manager. The default package manager for Centos is "yum."

```
TASK [install apache2 package] *****
[WARNING]: Updating cache and auto-installing missing dependency: python3-
fatal: [Galan_CentOS]: FAILED! => {"changed": false, "cmd": "apt-get update", "rc": 1, "stderr": "[Errno 2] No such file or directory: b'apt-get'", "stdout": ""}
ok: [server2]
ok: [server1]
```

3. Edit the `install_apache.yml` file and insert the lines shown below.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
GNU nano 7.2                                install_apache.yaml
- hosts: all
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2
        when: ansible_distribution == "Ubuntu"

    - name: update repository index
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        when: ansible_distribution == "Ubuntu"

[ Read 18 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Make sure to save the file and exit.

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

We will see in the play recap that it was able to reach all virtual machines in the 'Gathering Facts' phase, however we would observe that for each subsequent task, the Galan_CentOS VM was skipped because it does not fulfill the when condition clause.

If you have a mix of Debian and Ubuntu servers, you can change the configuration of your playbook like this.

```
- name: update repository index
  apt:
    update_cache: yes
    when: ansible_distribution in ["Debian", "Ubuntu"]
```

Note: This will work also if you try. Notice the changes are highlighted.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ ansible-playbook install_apache.y
aml -K -i inventory.yaml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
ok: [Galan_CentOS]

TASK [install apache2 package] *****
skipping: [Galan_CentOS]
ok: [server2]
ok: [server1]

TASK [update repository index] *****
skipping: [Galan_CentOS]
changed: [server2]
changed: [server1]

TASK [add PHP support for apache] *****
skipping: [Galan_CentOS]
ok: [server2]
ok: [server1]

PLAY RECAP *****
Galan_CentOS      : ok=1   changed=0   unreachable=0   failed=0   skip
ped=3   rescued=0   ignored=0
server1          : ok=4   changed=1   unreachable=0   failed=0   skip
ped=0   rescued=0   ignored=0
server2          : ok=4   changed=1   unreachable=0   failed=0   skip
ped=0   rescued=0   ignored=0
```

4. Edit the install_apache.yml file and insert the lines shown below.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
GNU nano 7.2 install_apache.yaml
- hosts: all
  become: true
  tasks:
    - name: install apache2 package
      apt:
        name: apache2
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: update repository index
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: update repository index
      dnf:
        update_cache: yes
        when: ansible_distribution == "CentOS"

    - name: install apache2 package
      dnf:
        name: httpd
        state: latest
        when: ansible_distribution == "CentOS"

    - name: add PHP support for apache
      dnf:
        name: php
        state: latest
        when: ansible_distribution == "CentOS"
```

5. To verify the installations, go to CentOS VM and type its IP address on the browser. Was it successful? The answer is no. It's because the httpd service or the Apache HTTP server in CentOS is not yet active. Thus, you need to activate it first.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
ok: [server1]

TASK [update repository index] *****
skipping: [Galan_CentOS]
changed: [server1]
changed: [server2]

TASK [add PHP support for apache] *****
skipping: [Galan_CentOS]
ok: [server1]
ok: [server2]

TASK [update repository index] *****
skipping: [server1]
skipping: [server2]
ok: [Galan_CentOS]

TASK [install apache2 package] *****
skipping: [server1]
skipping: [server2]
changed: [Galan_CentOS]

TASK [add PHP support for apache] *****
skipping: [server1]
skipping: [server2]
changed: [Galan_CentOS]

PLAY RECAP *****
Galan_CentOS      : ok=4    changed=2    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
server1      : ok=4    changed=1    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
server2      : ok=4    changed=1    unreachable=0    failed=0    skip
ped=3    rescued=0    ignored=0
```

5.1 To activate, go to the CentOS VM terminal and enter the following:
systemctl status httpd
The result of this command tells you that the service is inactive.

```
qimagalan@localhost:~
qimagalan-workstation@workstation: ~/CPE212... x qimagalan@localhost:~ x v
[qimagalan@localhost ~]$ systemctl status httpd
○ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: inactive (dead)
   Docs: man:httpd.service(8)
[qimagalan@localhost ~]$
```

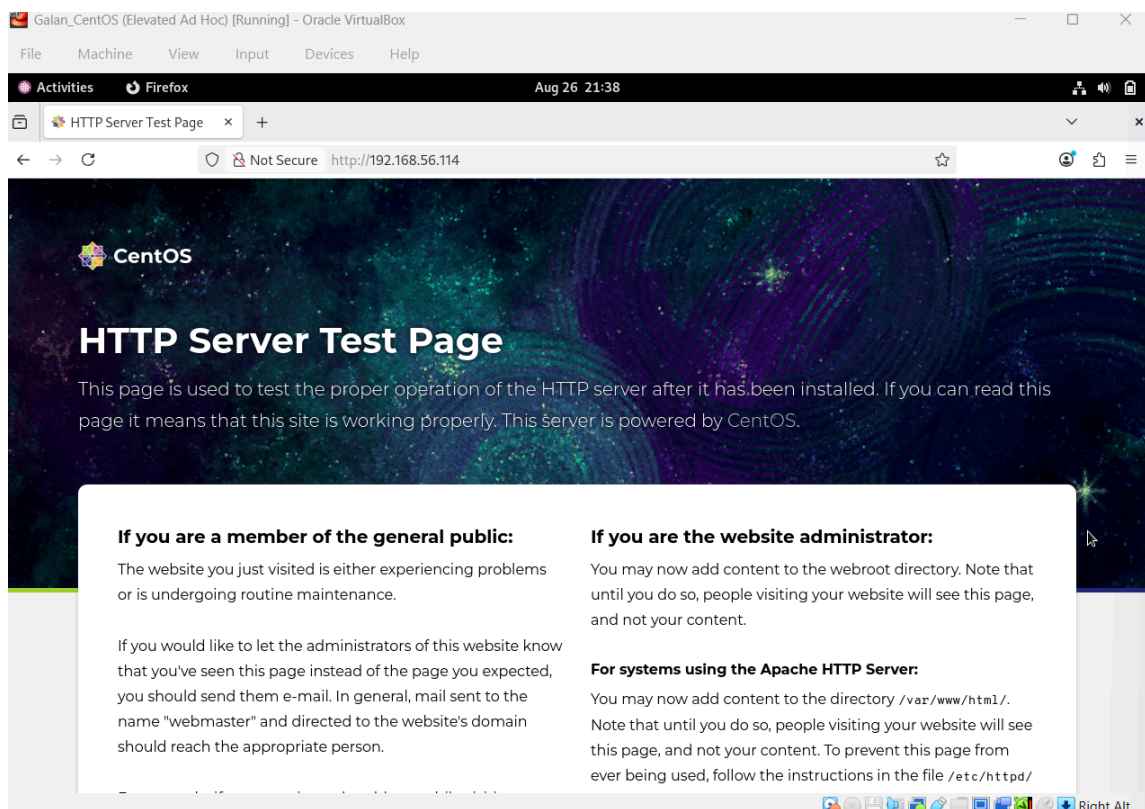
5.2 Issue the following command to start the service:
sudo systemctl start httpd (When prompted, enter the sudo password)
sudo firewall-cmd --add-port=80/tcp (The result should be a success)

```

qimagalan@localhost:~
qimagalan-workstation@workstation: ~/CPE212... x qimagalan@localhost:~

[qimagalan@localhost ~]$ sudo systemctl start httpd
[qimagalan@localhost ~]$ sudo firewall-cmd --add-port=80/tcp
success
[qimagalan@localhost ~]$
  
```

5.3 To verify the service is already running, go to CentOS VM and type its IP address on the browser. Was it successful? (Screenshot the browser)



Task 2: Refactoring playbook

This time, we want to make sure that our playbook is efficient and that the codes are easier to read. This will also makes run ansible more quickly if it has to execute fewer tasks to do the same thing.

1. Edit the playbook install_apache.yml. Currently, we have three tasks targeting our Ubuntu machines and 3 tasks targeting our CentOS machine. Right now, we try to consolidate some tasks that are typically the

same. For example, we can consolidate two plays that install packages. We can do that by creating a list of installation packages as shown below:

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x
GNU nano 7.2 install_apache.yaml
hosts: all
become: true
tasks:
- name: install apache2 and php packages for Ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    when: ansible_distribution == "Ubuntu"
- name: update repository index
  apt:
    update_cache: yes
    when: ansible_distribution == "Ubuntu"
- name: update repository index
  dnf:
    update_cache: yes
    when: ansible_distribution == "CentOS"
- name: install apache and php packages for CentOS
  dnf:
    name:
      - httpd
      - php
    state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the Result.

The refactored code already included the checking of `ansible_distribution` for CentOS, however it is quite inefficient because it totally separates the task of Ubuntu from CentOS, causing the play to skip CentOS on some tasks meanwhile the Ubuntu is also skipped for CentOS parts.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x
qimagalan-workstation@workstation:~/CPE212_Galan$ ansible-playbook install_apache.y
aml -i inventory.yaml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [Galan_CentOS]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [Galan_CentOS]
ok: [server2]
ok: [server1]

TASK [update repository index] *****
skipping: [Galan_CentOS]
changed: [server2]
changed: [server1]

TASK [update repository index] *****
skipping: [server1]
skipping: [server2]
ok: [Galan_CentOS]

TASK [install apache and php packages for CentOS] *****
skipping: [server1]
skipping: [server2]
ok: [Galan_CentOS]

PLAY RECAP *****
Galan_CentOS      : ok=3    changed=0    unreachable=0    failed=0    skip
ped=2    rescued=0    ignored=0
server1          : ok=3    changed=1    unreachable=0    failed=0    skip
```

2. Edit the playbook install_apache.yml again. In task 2.1, we consolidated the plays into one play. This time we can actually consolidate everything in just 2 plays. This can be done by removing the update repository play and putting the command update_cache: yes below the command state: latest. See below for reference:

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE2
GNU nano 7.2 install_apache.yaml
- hosts: all
  become: true
  tasks:
    - name: install apache2 and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
        when: ansible_distribution == "Ubuntu"
    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
        when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

This refactored form is much more efficient than the first one because it consolidates the package installations and cache updates. Instead of skipping a non-matching distribution once per task, this form reduced it to just one skip only because there is now only one task for each distro.

```

qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x
qimagalan-workstation@workstation:~/CPE212_Galan$ ansible-playbook install_apache.y
aml -i inventory.yaml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [Galan_CentOS]

TASK [install apache2 and php packages for Ubuntu] *****
skipping: [Galan_CentOS]
ok: [server2]
ok: [server1]

TASK [install apache and php packages for CentOS] *****
skipping: [server1]
skipping: [server2]
ok: [Galan_CentOS]

PLAY RECAP *****
Galan_CentOS      : ok=2   changed=0   unreachable=0   failed=0   skip
ped=1   rescued=0   ignored=0
server1          : ok=2   changed=0   unreachable=0   failed=0   skip
ped=1   rescued=0   ignored=0
server2          : ok=2   changed=0   unreachable=0   failed=0   skip
ped=1   rescued=0   ignored=0
qimagalan-workstation@workstation:~/CPE212_Galan$

```

3. Finally, we can consolidate these 2 plays in just 1 play. This can be done by declaring variables that will represent the packages that we want to install. Basically, the `apache_package` and `php_package` are variables. The names are arbitrary, which means we can choose different names. We also take out the line `when: ansible_distribution`. Edit the playbook `install_apache.yml` again and make sure to follow the below image. Make sure to save the file and exit.

```

qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation:
GNU nano 7.2                                install_apache.yaml
- hosts: all
  become: true
  tasks:
    - name: install apache and php
      apt:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
#   when: ansible_distribution == "Ubuntu"

```

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

This part is meant to fail because we have defined variables that are nowhere to be found in the inventory file that we are using. Namely, these variables are `apache_package` and `php_package`.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x v
PLAY [all] *****
TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
ok: [Galan_CentOS]

TASK [install apache and php] *****
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/qimagalan-workstation/CPE212_Galan/install_apache.yaml': line 4, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n tasks:\n - name: install apache and php\n   ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/qimagalan-workstation/CPE212_Galan/install_apache.yaml': line 4, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n tasks:\n - name: install apache and php\n   ^ here\n"}
fatal: [Galan_CentOS]: FAILED! => {"msg": "The task includes an option with an undefined variable. The error was: 'apache_package' is undefined. 'apache_package' is undefined\n\nThe error appears to be in '/home/qimagalan-workstation/CPE212_Galan/install_apache.yaml': line 4, column 5, but may\nbe elsewhere in the file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n tasks:\n - name: install apache and php\n   ^ here\n"}

PLAY RECAP *****
Galan_CentOS      : ok=1    changed=0    unreachable=0    failed=1    skip
ped=0    rescued=0    ignored=0
server1          : ok=1    changed=0    unreachable=0    failed=1    skip
ped=0    rescued=0    ignored=0
server2          : ok=1    changed=0    unreachable=0    failed=1    skip
ped=0    rescued=0    ignored=0
```

- Unfortunately, task 2.3 was not successful. It's because we need to change something in the inventory file so that the variables we declared will be in place. Edit the inventory file and follow the below configuration:

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x
GNU nano 7.2 inventory.yaml
all:
  children:
    Ubuntu:
      vars:
        OpSys: Ubuntu
        apache_package: apache2
        php_package: libapache2-mod-php
      hosts:
        server1:
          ansible_host: 192.168.56.113
        server2:
          ansible_host: 192.168.56.106

    CentOSes:
      vars:
        ansible_user: qimagalan
        OpSys: CentOS
        apache_package: httpd
        php_package: php
      hosts:
        Galan_CentOS:
          ansible_host: 192.168.56.114

    VMs:
      children:
        Ubuntu:
        CentOSes:
```

Make sure to save the inventory file and exit.

Finally, we still have one more thing to change in our `install_apache.yml` file. In task 2.3, you may notice that the package is assigned as `apt`, which will not run in CentOS. Replace the `apt` with the `package`. `Package` is a module in ansible that is generic, which is going to use whatever package manager the underlying host or the target server uses. For Ubuntu it will automatically use `apt`, and for CentOS it will automatically use `dnf`. Make sure to save the file and exit. For more details about the ansible package, you may refer to this documentation: [ansible.builtin.package – Generic OS package manager — Ansible Documentation](#)

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation:
GNU nano 7.2 install_apache.yaml
- hosts: all
  become: true
  tasks:
    - name: install apache and php
      package:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

Run `ansible-playbook --ask-become-pass install_apache.yml` and describe the result.

This task will now succeed as I have modified my inventory.yaml file in the screenshot above. However, I used the modern syntax for yaml instead of the syntax we often use for the default /etc/ansible/hosts. I simply declared the apache_package and php_package on the vars section directly under the metagroup in order to make it scalable as we add more machines in the future if ever.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212... x qimagalan-workstation@workstation: ~/CPE212... x v
qimagalan-workstation@workstation:~/CPE212_Galan$ ansible-playbook install_apache.y
aml -i inventory.yaml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [Galan_CentOS]

TASK [install apache and php] *****
ok: [server2]
ok: [server1]
ok: [Galan_CentOS]

PLAY RECAP *****
Galan_CentOS      : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0
server1          : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0
server2          : ok=2    changed=0    unreachable=0    failed=0    skip
ped=0    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan$
```

Saving a copy to github:

```

qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation: ~/CPE212_Galan$ git add install_apache_HOA5.yaml
inventory.yaml
qimagalan-workstation@workstation: ~/CPE212_Galan$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   install_apache_HOA5.yaml
        new file:   inventory.yaml

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   install_apache.yaml

qimagalan-workstation@workstation: ~/CPE212_Galan$ git commit -m "HOA 5 Added"
[main d839a35] HOA 5 Added
 2 files changed, 37 insertions(+)
 create mode 100644 install_apache_HOA5.yaml
 create mode 100644 inventory.yaml
qimagalan-workstation@workstation: ~/CPE212_Galan$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   install_apache.yaml

qimagalan-workstation@workstation: ~/CPE212_Galan$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 716 bytes | 238.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:IrickMarvinGalan/CPE212_Galan.git
   f759998..d839a35  main -> main
qimagalan-workstation@workstation: ~/CPE212_Galan$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   install_apache.yaml

no changes added to commit (use "git add" and/or "git commit -a")

```

	IrickMarvinGalan	HOA 5 Added	d839a35 · 2 minutes ago	
	README.md	README updated	last month	
	install_apache.yaml	Install Apache Playbook	last week	
	install_apache_HOA5.yaml	HOA 5 Added	2 minutes ago	
	inventory.yaml	HOA 5 Added	2 minutes ago	
	playbook.yaml	Commit Playbook	last week	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:

Create a playbook that could do the previous tasks in Red Hat OS.

- 1.) Run `sudo dnf install python3-pip` to resolve ansible's dependency on it.

```

qimagalan@GalanCentOS:~
[qimagalan@GalanCentOS ~]$ sudo dnf install ansible
[sudo] password for qimagalan:
CentOS Stream 9 - BaseOS                    595 B/s | 3.4 kB    00:05
CentOS Stream 9 - BaseOS                    1.2 MB/s | 8.8 MB   00:07
CentOS Stream 9 - AppStream                 3.9 kB/s | 3.1 kB    00:00
CentOS Stream 9 - AppStream                 1.9 MB/s | 28 MB    00:14
CentOS Stream 9 - Extras packages           3.1 kB/s | 5.0 kB    00:01
CentOS Stream 9 - Extras packages           5.3 kB/s | 22 kB     00:04
Last metadata expiration check: 0:00:01 ago on Wed 26 Aug 2026 08:42:11 PM EDT.
No match for argument: ansible
Error: Unable to find a match: ansible
[qimagalan@GalanCentOS ~]$

```

```
qimagalan@GalanCentOS:~
[qimagalan@GalanCentOS ~]$ sudo dnf install python3-pip
Last metadata expiration check: 0:02:06 ago on Wed 26 Aug 2026 08:42:11 PM EDT.
Dependencies resolved.
=====
Package                Architecture Version                Repository            Size
=====
Installing:
python3-pip            noarch             21.3.1-2.el9          appstream             2.0 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 2.0 M
Installed size: 8.8 M
Is this ok [y/N]: Y
Downloading Packages:
[MIRROR] python3-pip-21.3.1-2.el9.noarch.rpm: Curl error (28): Timeout was reach
ed for http://mirrors.tuna.tsinghua.edu.cn/centos-stream/9-stream/AppStream/x86_
64/os/Packages/python3-pip-21.3.1-2.el9.noarch.rpm [Failed to connect to mirrors
.tuna.tsinghua.edu.cn port 80: Connection timed out]
python3-pip-21.3.1-2.el9.noarch.rpm                49 kB/s | 2.0 MB    00:42
=====
Total                                                47 kB/s | 2.0 MB    00:44
```

- 2.) Run `sudo dnf update -y` to update the system before installing ansible.

```
qimagalan@GalanCentOS:~ — sudo dnf install ansible
qimagalan@GalanCentOS:~ — su... x qimagalan@GalanCentOS:~ x
python3-pip-21.3.1-2.el9.noarch
Complete!
[qimagalan@GalanCentOS ~]$ sudo dnf update -y
Last metadata expiration check: 0:04:32 ago on Wed 26 Aug 2026 08:42:11 PM EDT.
Dependencies resolved.
=====
Package                Arch    Version                Repo    Size
=====
Installing:
kernel                 x86_64  5.14.0-741.el9        baseos  785 k
Upgrading:
NetworkManager         x86_64  1:1.54.4-3.el9        baseos  2.4 M
NetworkManager-adsl    x86_64  1:1.54.4-3.el9        baseos  29 k
NetworkManager-bluetooth x86_64  1:1.54.4-3.el9        baseos  55 k
NetworkManager-config-server noarch  1:1.54.4-3.el9        baseos  15 k
NetworkManager-libnm   x86_64  1:1.54.4-3.el9        baseos  1.9 M
NetworkManager-team    x86_64  1:1.54.4-3.el9        baseos  34 k
NetworkManager-tui     x86_64  1:1.54.4-3.el9        baseos  245 k
NetworkManager-wifi    x86_64  1:1.54.4-3.el9        baseos  78 k
NetworkManager-wwan    x86_64  1:1.54.4-3.el9        baseos  62 k
acl                    x86_64  2.4.0-2.el9           baseos  76 k
audit                  x86_64  3.1.5-9.el9           baseos  273 k
audit-libs             x86_64  3.1.5-9.el9           baseos  120 k
```

- 3.) Since ansible could not be found by `dnf install`, we still have to run `sudo dnf install epel-release`. We should be able to find and install ansible after.

```
qimagalan@GalanCentOS:~ — sudo dnf install ansible
```

```
qimagalan@GalanCentOS:~ — su... x qimagalan@GalanCentOS:~ x
```

```
Error: Unable to find a match: ansible
[qimagalan@GalanCentOS ~]$ sudo dnf install epel-release -y
Last metadata expiration check: 0:01:32 ago on Wed 26 Aug 2026 09:08:24 PM EDT.
Dependencies resolved.
=====
Package                Architecture Version      Repository      Size
=====
Installing:
  epel-release          noarch      9-7.el9        extras-common   19 k
Installing weak dependencies:
  epel-next-release     noarch      9-7.el9        extras-common   8.1 k

Transaction Summary
=====
Install 2 Packages

Total download size: 27 k
Installed size: 29 k
Downloading Packages:
(1/2): epel-next-release-9-7.el9.noarch.rpm    45 kB/s | 8.1 kB    00:00
(2/2): epel-release-9-7.el9.noarch.rpm        95 kB/s | 19 kB    00:00
-----
Total                                           27 kB/s | 27 kB    00:00
CentOS Stream 9 - Extras packages             911 kB/s | 2.1 kB  00:00
```

- 4.) We run `sudo dnf install ansible` to finally install the ansible in our CentOS control node.

```
qimagalan@GalanCentOS:~ — sudo dnf install ansible
```

```
qimagalan@GalanCentOS:~ — su... x qimagalan@GalanCentOS:~ x
```

```
Complete!
[qimagalan@GalanCentOS ~]$ sudo dnf install ansible
Extra Packages for Enterprise Linux 9 - x86_64 1.2 MB/s | 20 MB 00:16
Extra Packages for Enterprise Linux 9 openh264 836 B/s | 2.5 kB 00:03
Extra Packages for Enterprise Linux 9 - Next - 127 kB/s | 265 kB 00:02
Last metadata expiration check: 0:00:01 ago on Wed 26 Aug 2026 09:12:41 PM EDT.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
ansible noarch 1:7.7.0-1.el9 epel 34 M
Installing dependencies:
ansible-core x86_64 1:2.14.18-5.el9 appstream 2.6 M
git-core x86_64 2.52.0-1.el9 appstream 5.0 M
python3-cffi x86_64 1.14.5-5.el9 baseos 253 k
python3-cryptography x86_64 36.0.1-5.el9 baseos 1.2 M
python3-packaging noarch 20.9-5.el9 appstream 77 k
python3-ply noarch 3.11-14.el9 baseos 106 k
python3-pycparser noarch 2.20-6.el9 baseos 135 k
python3-pyparsing noarch 2.4.7-9.el9 baseos 150 k
python3-resolvelib noarch 0.5.4-5.el9 appstream 34 k
sshpas x86_64 1.09-4.el9 appstream 28 k
```

- 5.) After installing ansible, we still need to run *sudo dnf install git* as we have to clone and update our repository as we progress in future activities.

```
qimagalan@GalanCentOS:~ — sudo dnf install git
```

```
qimagalan@GalanCentOS:~ — su... x qimagalan@GalanCentOS:~ x
```

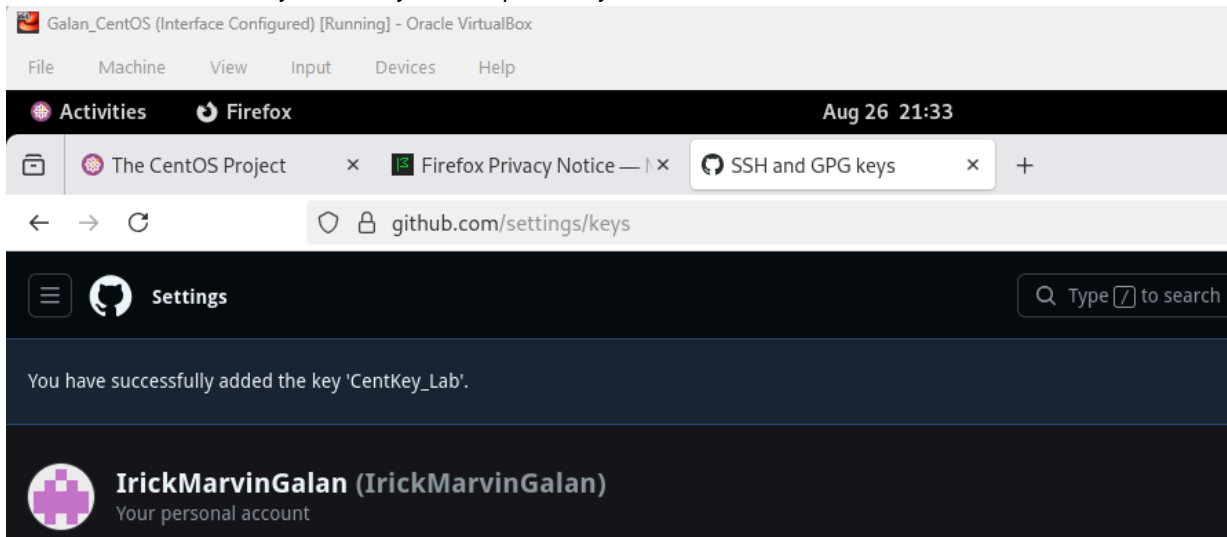
```
Complete!
[qimagalan@GalanCentOS ~]$ sudo dnf install git
[sudo] password for qimagalan:
Last metadata expiration check: 0:08:09 ago on Wed 26 Aug 2026 09:12:41 PM EDT.
Dependencies resolved.
=====
Package                Arch      Version      Repository    Size
=====
Installing:
git                    x86_64    2.52.0-1.el9 appstream     39 k
Installing dependencies:
git-core-doc           noarch    2.52.0-1.el9 appstream     3.1 M
perl-Error              noarch    1:0.17029-7.el9 appstream     42 k
perl-Git                noarch    2.52.0-1.el9 appstream     37 k
perl-TermReadKey        x86_64    2.38-11.el9  appstream     37 k
perl-lib               x86_64    0.65-483.el9 appstream     15 k
=====
Transaction Summary
=====
Install 6 Packages

Total download size: 3.3 M
Installed size: 19 M
```

- 6.) We need to setup ssh before we could begin cloning, pulling, and pushing to the git repository. The first step is to run `ssh-keygen -t <ssh key type> -f <output file>`. In this case I used `ed25519` since it is smaller and a little faster than `rsa`.

```
qimagalan@GalanCentOS:~
qimagalan@GalanCentOS:~ x qimagalan@GalanCentOS:~ x
[qimagalan@GalanCentOS ~]$ ssh-keygen -t ed25519 -f ~/.ssh/GitKey_Cent
Generating public/private ed25519 key pair.
Enter passphrase for "/home/qimagalan/.ssh/GitKey_Cent" (empty for no passphrase
):
Enter same passphrase again:
Your identification has been saved in /home/qimagalan/.ssh/GitKey_Cent
Your public key has been saved in /home/qimagalan/.ssh/GitKey_Cent.pub
The key fingerprint is:
SHA256:P3ejMpzhMjTWrfLzfylFWTPp23e7lTUKCp19ERZWSRs qimagalan@GalanCentOS
The key's randomart image is:
+---[ED25519 256]---+
|      .EB|
|      . o o==|
|      . o o.+=|
|      . . ++|
|      S. o ..+|
|      +.o . oB|
|      o +o+. +.*|
|      + 0o + +o|
|      =.=o.oo.|
+-----[SHA256]-----+
[qimagalan@GalanCentOS ~]$
```

- 7.) Although not shown in this step, we have to run `sudo cat ~/.ssh/<output file name>.pub` to get the public key that we would copy in the SSH and GPG keys in our github account. Below is shown that I successfully added the CentKey_Lab in my allowed public keys.



- 8.) Next we configure the git config for username and email, because it is a necessary part of the setup.

```
qimagalan@GalanCentOS:~
[qimagalan@GalanCentOS ~]$ git config --global user.name "Irick Marvin Galan"
[qimagalan@GalanCentOS ~]$ git config --global user.email "qimagalan@tip.edu.ph"
[qimagalan@GalanCentOS ~]$
```

9.) After setting up credentials, we are ready to run *git clone <github link>*

```
qimagalan@GalanCentOS:~
[qimagalan@GalanCentOS ~]$ git clone git@github.com:IrickMarvinGalan/CPE212_Galan.git
Cloning into 'CPE212_Galan'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 34, done.
remote: Counting objects: 100% (34/34), done.
remote: Compressing objects: 100% (28/28), done.
remote: Total 34 (delta 7), reused 19 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (34/34), 6.33 KiB | 3.16 MiB/s, done.
Resolving deltas: 100% (7/7), done.
[qimagalan@GalanCentOS ~]$
```

10.) We have to do *ssh-copy-id -i <path to pubKey> user@server* since the CentOS is a new control node that needs its public key added to the *authorized_keys* file of remote servers.

```

qimagalan@GalanCentOS:~/CPE212_Galan
[qimagalan@GalanCentOS CPE212_Galan]$ ssh-copy-id -i ~/.ssh/GitKey_Cent.pub qimagalan@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/qimagalan/.ssh/GitKey_Cent.pub"
The authenticity of host 'server1 (192.168.56.127)' can't be established.
ED25519 key fingerprint is SHA256:vlXUPJ5JeqdliTTntFHQ1JUvL4l95k09DjFt2H4zMTU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
qimagalan@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/qimagalan/.ssh/GitKey_Cent 'qimagalan@server1'"
and check to make sure that only the key(s) you wanted were added.

[qimagalan@GalanCentOS CPE212_Galan]$ ssh-copy-id -i ~/.ssh/GitKey_Cent.pub qimagalan@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/qimagalan/.ssh/GitKey_Cent.pub"
The authenticity of host 'server2 (192.168.56.128)' can't be established.
ED25519 key fingerprint is SHA256:vlXUPJ5JeqdliTTntFHQ1JUvL4l95k09DjFt2H4zMTU.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: server1
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
qimagalan@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/qimagalan/.ssh/GitKey_Cent 'qimagalan@server2'"
and check to make sure that only the key(s) you wanted were added.

Galan_CentOS (SSH and Git Setup) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Activities Terminal Aug 26 22:04

qimagalan@GalanCentOS:~/CPE212_Galan
[qimagalan@GalanCentOS CPE212_Galan]$ sudo nano /etc/hosts
[qimagalan@GalanCentOS CPE212_Galan]$ ssh-copy-id -i ~/.ssh/GitKey_Cent qimagalan@workstation
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/qimagalan/.ssh/GitKey_Cent.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
>/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
qimagalan@workstation's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/qimagalan/.ssh/GitKey_Cent 'qimagalan@workstation'"
and check to make sure that only the key(s) you wanted were added.

[qimagalan@GalanCentOS CPE212_Galan]$

```

- 11.) Finally we could run the *ansible-playbook install_apache.yaml -i inventory.yaml -K* to check if the configurations so far do work. In this case, my file is named *install_apache_HOA5.yaml* to prepare for staging, commit, and pushing to the github repository.

```

qimagalan@GalanCentOS:~/CPE212_Galan
[qimagalan@GalanCentOS CPE212_Galan]$ ansible-playbook install_apache_H0A5.yaml -i inventory.yaml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [workstation]

TASK [install apache and php] *****
ok: [server2]
ok: [server1]
changed: [workstation]

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
workstation  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0

[qimagalan@GalanCentOS CPE212_Galan]$

```

Reflections:

1. Why do you think refactoring of playbook codes is important?

In the first part of the procedure we observed that the task execution is tedious and more elaborate than necessary. It is able to accomplish the job, but its readability is not clear because similar logic are not grouped together but executed one by one on their own. This style of creating playbooks is valid for prototyping, trying to come up with a straightforward solution, but is bound to become highly unscalable as a project progresses further and further into development. Bugs become much more commonplace when the ansible playbook is more verbose.

2. When do we use the “when” command in the playbook?

The when command becomes relevant when checking a specific condition before carrying out a specific task is necessary. In this activity, we observed how it could be applied by checking a builtin variable called `ansible_distribution` that retrieves the Linux Distro utilized by a specific Virtual Machine.

Conclusion

In this activity, I learned another fundamental skill in building ansible playbooks which is primarily the use of the when command for conditional executions. I applied the `ansible_distribution` to check the Linux Distro used by a managed node and adaptively use `apt` or `dnf` depending on the specific distro. Towards the final version of the refactoring part, I also learned that using when for package installation is unnecessary as there exists a module for package that automatically distinguishes between the Distros. In the future, I will search for modules that may become adaptable to make refactoring easier, so all that I have to do left is to reconcile the discrepancies in names or some basic information by declaring them in vars block in the inventory file.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:



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No AI was used for this activity

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No AI was used for this activity

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